上海科技大学
ShanghaiTech University

CS240 Algorithm Design and Analysis

Lecture 15

PSPACE: A Class of Problems Beyond NP

Qingying Hao
Spring 2026
2026.04.16

Geography Game

- **Geography.** Amy names capital city c of country she is in. Bob names a capital city c' that starts with the letter on which c ends. Amy and Bob repeat this game until one player is unable to continue. Does Amy have a forced win?
- **Ex.** Budapest → Tokyo → Ottawa → Ankara → Amsterdam → Moscow → Washington → Nairobi → …

- **Geography on graphs.** Given a directed graph G = (V, E) and a start node s, two players alternate turns by following, if possible, an edge out of the current node to an unvisited node. Can first player guarantee to make the last legal move?

- **Remark.** Some problems (especially involving 2-player games and AI) defy classification according to P, NP, and NP-complete

PSPACE

- **P.** Decision problems solvable in polynomial time
- **PSPACE.** Decision problems solvable in polynomial space

- **Observation.** P ⊆ PSPACE

Poly-time algorithm can consume only polynomial space

PSPACE

- **Claim.** 3-SAT is in PSPACE
- **Pf.**
  - Enumerate all 2^n possible truth assignments
  - Check each assignment to see if it satisfies all clauses

- **Theorem.** NP ⊆ PSPACE
- **Pf.** Consider arbitrary problem Y in NP
  - Since Y $\leq_p$ 3-SAT, there exists algorithm that solves Y in poly-time plus polynomial number of calls to 3-SAT black box
  - Can implement black box in poly-space

Quantified Satisfiability

- **QSAT (Quantified 3-SAT).** Let Φ(X1,…Xn) be a Boolean CNF formula. Is the following propositional formula true?

$$\exists x_1 \, \forall x_2 \, \exists x_3 \, \forall x_4 \ldots \forall x_{n-1} \, \exists x_n \; \Phi(x_1, \ldots, x_n)$$

assume n is odd

- **Intuition.** Amy picks truth value for X_1, then Bob for X_2, then Amy for X_3, and so on. Can Amy satisfy Φ no matter what Bob does?

- **Ex.** $(x_1 \vee x_2) \wedge (x_2 \vee \overline{x_3}) \wedge (\overline{x_1} \vee \overline{x_2} \vee x_3)$ $\quad \exists x_1 \, \forall x_2 \, \exists x_3 \; \Phi(x_1, x_2, x_3)$

  Yes. Amy sets X_1 true; Bob sets X_2; Amy sets X_3 to be same as X_2

- **Ex.** $(x_1 \vee x_2) \wedge (\overline{x_2} \vee \overline{x_3}) \wedge (\overline{x_1} \vee \overline{x_2} \vee x_3)$

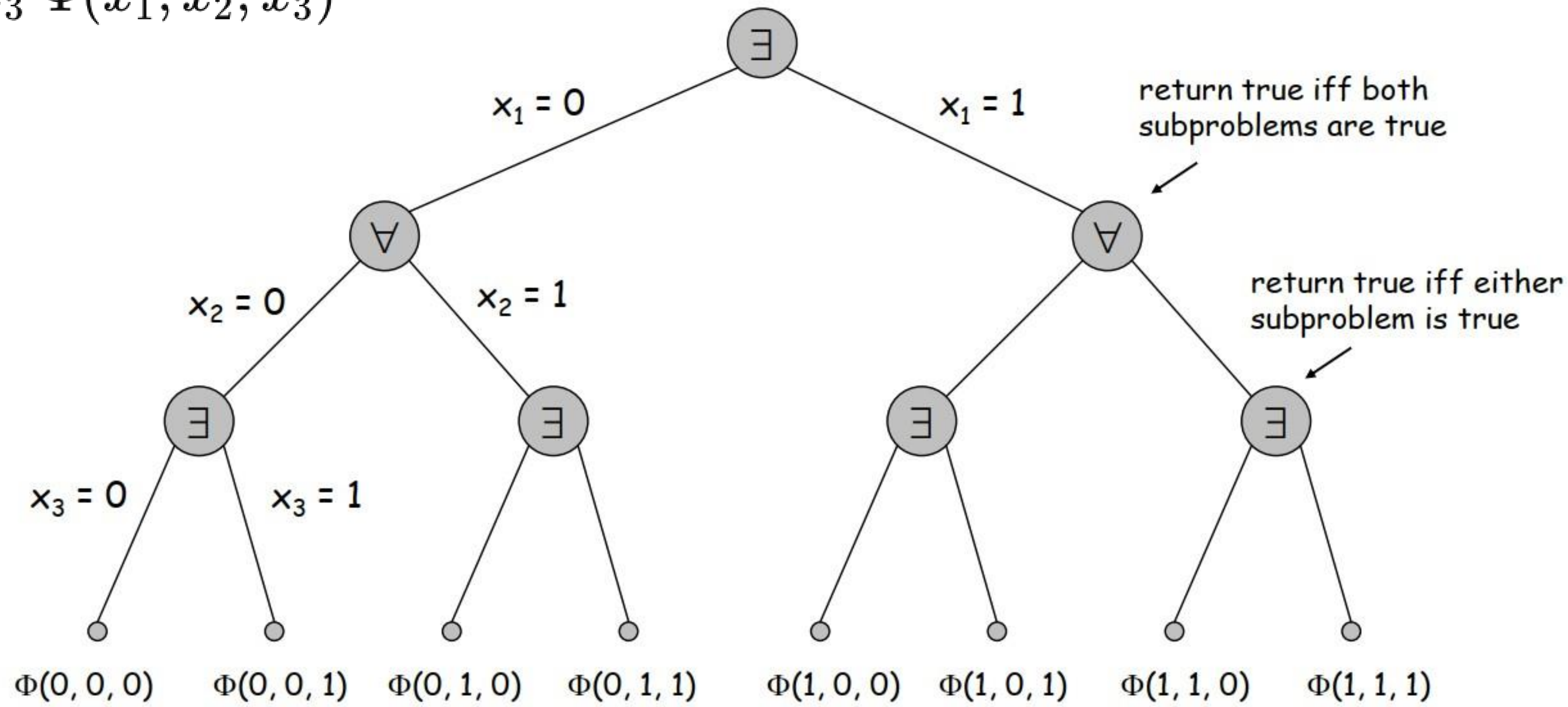
  No. If Amy sets X_1 false; Bob sets X_2 false; Amy loses;

  If Amy sets X_1 true; Bob sets X_2 true; Amy loses

QSAT is in PSPACE

- **Theorem.** QSAT ∈ PSPACE
- **Pf.** Recursively try all possibilities
  - Only need one bit of information from each subproblem
  - Amount of space is proportional to depth of function call stack

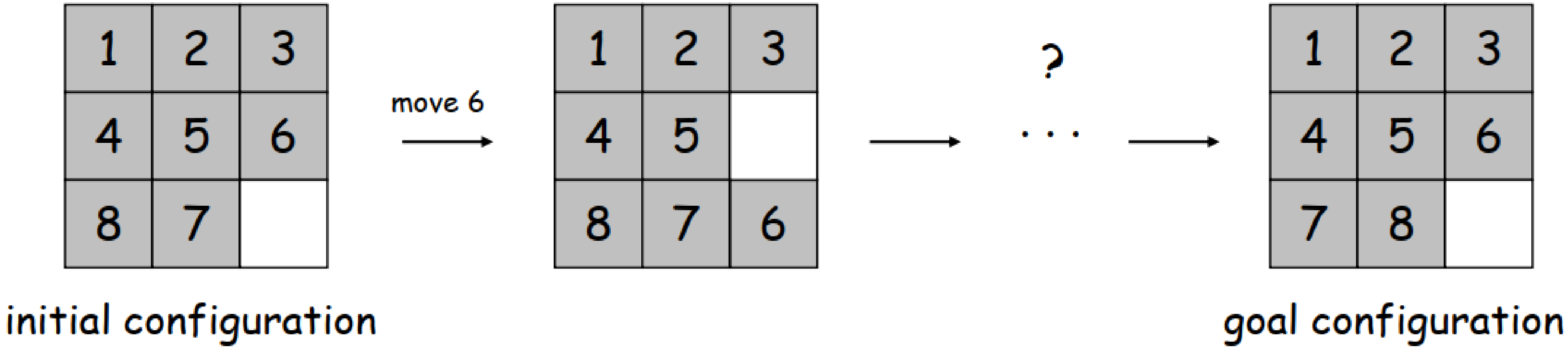
$$\exists x_1 \, \forall x_2 \, \exists x_3 \, \Phi(x_1, x_2, x_3)$$



Planning Problem

15-Puzzle

- **8-puzzle, 15-puzzle.** [Sam Loyd 1870s]
  - Board: 3-by-3 grid of tiles labeled 1-8
  - Legal move: slide neighboring tile into blank (white) square
  - Find sequence of legal moves to transform initial configuration into goal configuration



Planning Problem

- **Conditions.** Set $C = \{ C_1, \ldots, C_n \}$
- **Initial configuration.** Subset $c_0 \subseteq C$ of conditions initially satisfied
- **Goal configuration.** Subset $c^* \subseteq C$ of conditions we seek to satisfy
- **Operations.** Set $O = \{ O_1, \ldots, O_k \}$
  - To invoke operator Oi, must satisfy certain prereq conditions
  - After invoking Oi certain conditions become true, and certain conditions become false

- **PLANNING.** Is it possible to apply sequence of operators to get from initial configuration to goal configuration?

- **Examples.**
- 15-puzzle
- Rubik's cube
- Logistical operations to move people, equipment, and materials

Planning Problem: 8-Puzzle

- **Planning example.** Can we solve the 8-puzzle?
- **Conditions.** $C_{ij}, 1 \le i, j \le 9.$ $\leftarrow$ C_{ij} means tile i is in square j
- **Initial state.** $c_0 = \{C_{11}, C_{22}, \ldots, C_{66}, C_{78}, C_{87}, C_{99}\}.$
- **Goal state.** $c^* = \{C_{11}, C_{22}, \ldots, C_{66}, C_{77}, C_{88}, C_{99}\}.$
- **Operators.**
  - Precondition to apply $O_i = \{ C_{78}, C_{99} \}$
  - After invoking O_i, conditions C_{79} and C_{98} become true
  - After invoking O_i, conditions C_{78} and C_{99} become false

1	2	3
4	5	6
8	7	9

$\downarrow O_i$

1	2	3
4	5	6
8	9	7

Planning Problem: In Exponential Time

- **Configuration graph G**
  - Include node for each of 2^n possible configurations
  - Include an edge from configuration c' to configuration c'' if one of the operators can convert from c' to c''

- **PLANNING.** Is there a path from c_0 to c* in configuration graph?

- **Claim.** PLANNING is in EXPTIME
- **Pf.** Run BFS to find path from c_0 to c* in configuration graph

- **Note.** Configuration graph can have 2^n nodes, and shortest path can be of length $= 2^n - 1$

Planning Problem: In Polynomial Space

- **Theorem.** PLANNING is in PSPACE
- Pf.
  - Suppose there is a path from c_1 to c_2 of length $\leq L$
  - Path from c_1 to midpoint and from midpoint to c_2 are each $\leq L/2$
  - Enumerate all possible midpoints
  - Apply recursively. Depth of recursion $= \log_2 L$

```
boolean hasPath(c1, c2, L) {
   if (L = 1) return correct answer

   foreach configuration c' {
      boolean x = hasPath(c1, c', ⌈L/2⌉)
      boolean y = hasPath(c', c2, ⌊L/2⌋)
      if (x and y) return true
   }
   return false
}
```

PSPACE-Complete

PSPACE-Complete

- **PSPACE.** Decision problems solvable in polynomial space
- **PSPACE-Complete.** Problem Y is PSPACE-complete if (i) Y is in PSPACE and (ii) for every problem X in PSPACE, $X \leq_p Y$

- **Theorem.** [Stockmeyer-Meyer 1973] QSAT is PSPACE-complete
- **Theorem.** PSPACE $\subseteq$ EXPTIME
- **Pf.** Previous algorithm solves QSAT in exponential time, and QSAT is PSPACE-complete

- **Summary.** P $\subseteq$ NP $\subseteq$ PSPACE $\subseteq$ EXPTIME

it is known that P $\neq$ EXPTIME, but unknown which inclusion is strict; conjectured that all are

PSPACE-Complete Problems

上海科技大学 ShanghaiTech University

- More PSPACE-complete problems
  - **Competitive facility location**
  - Natural generalizations of games
    - Othello, Hex, Geography, Rush-Hour, Instant Insanity
    - Shanghai, go-moku, Sokoban
  - Given a memory restricted Turing machine, does it terminate in at most k steps?
  - Do two regular expressions describe different languages?
  - Is it possible to move and rotate complicated object with attachments through an irregularly shaped corridor?
  - Is a deadlock state possible within a system of communicating processors?

立志成才报国裕民

Competitive Facility Location

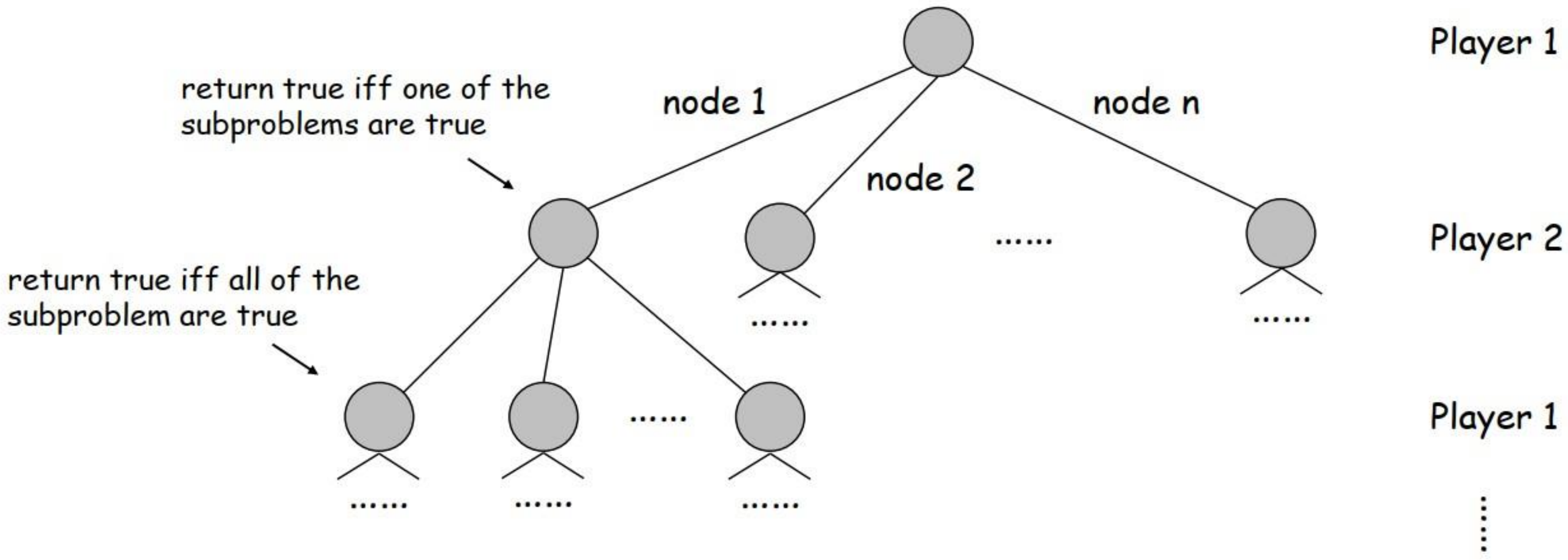
- **Input.** Graph with positive node weights, and target number B
- **Game.** Two competing players alternate in selecting nodes. Not allowed to select a node if any of its neighbors has been selected
- **Competitive facility location.** Can second player guarantee at least B units of profit?

(In general, the graph may not be a chain.)

Yes if B = 20; no if B = 25.

Competitive Facility Location

- **Claim.** COMPETITIVE-FACILITY is PSPACE-complete
- Pf.
- To solve it in poly-space, use recursion like QSAT, but at each step there are up to n choices instead of 2



Competitive Facility Location

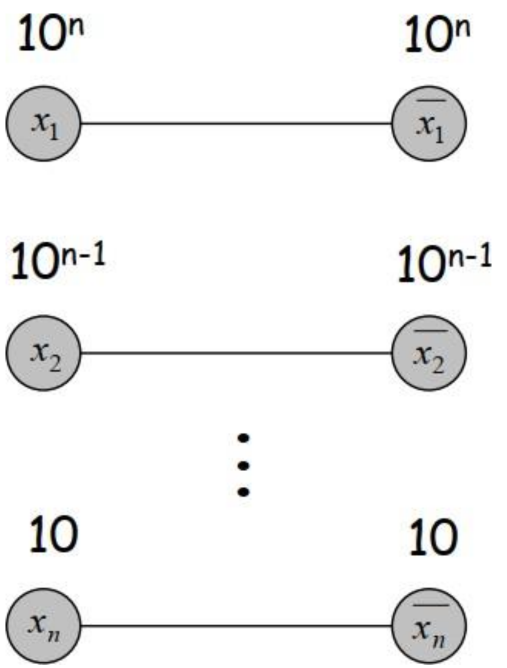
- **Claim.** COMPETITIVE-FACILITY is PSPACE-complete
- **Pf.**
  - To show that it's complete, we show that QSAT polynomial reduces to it.
  - Given an instance of QSAT, we construct an instance of COMPETITIVE-FACILITY such that player 2 can force a win iff QSAT formula is **false**

Competitive Facility Location

上海科技大学 ShanghaiTech University

Assume n is odd

- **Construction.** Given instance $\Phi(X1, \ldots, Xn) = C1 \wedge C2 \wedge \ldots Ck$ of QSAT
  - Include a node for each literal and its negation and connect them
    - At most one of X_i and its negation can be chosen
  - Choose a large constant c (e.g., $c \geq k+2$), and put weight c^{n-i+1} on literal x_i and its negation;
  - Set $B = c^{n-1} + c^{n-3} + \ldots + c^4 + c^2 + 1$
    - This ensures variables are selected in order $x_1, x_2, \ldots, x_n$
  - As is, player 2 will lose by 1 unit: $c^{n-1} + c^{n-3} + \ldots + c^4 + c^2$

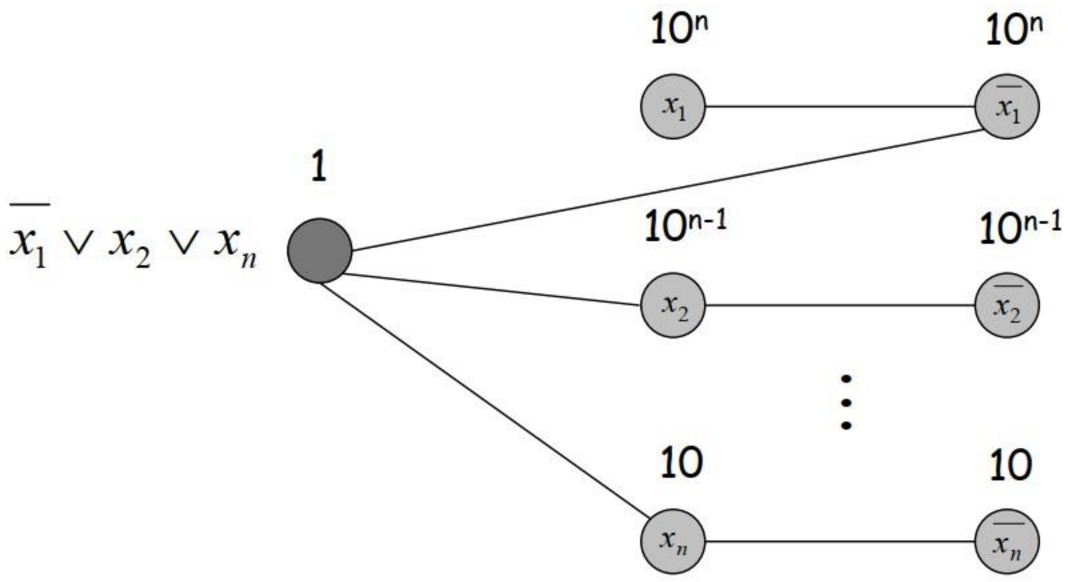


立志成才报国裕民

Competitive Facility Location

- **Construction.** Given instance $\Phi(X_1, \ldots, X_n) = C_1 \wedge C_2 \wedge \ldots C_k$ of QSAT
  - Given player 2 one last move on which she can try to win
  - For each clause C_j, add node with value 1 and an edge to each of its literals
  - Player 2 can make last move iff truth assignment defined alternately by the players failed to satisfy some clause, i.e.,

$$\forall x_1\ \exists x_2\ \forall x_3\ \exists x_4 \ldots \exists x_{n-1}\ \forall x_n\ \neg\Phi(x_1, \ldots, x_n)$$
$$\Leftrightarrow\ \neg\ \exists x_1\ \forall x_2\ \exists x_3\ \forall x_4 \ldots\ \forall x_{n-1}\ \exists x_n\ \Phi(x_1, \ldots, x_n)$$



Summary

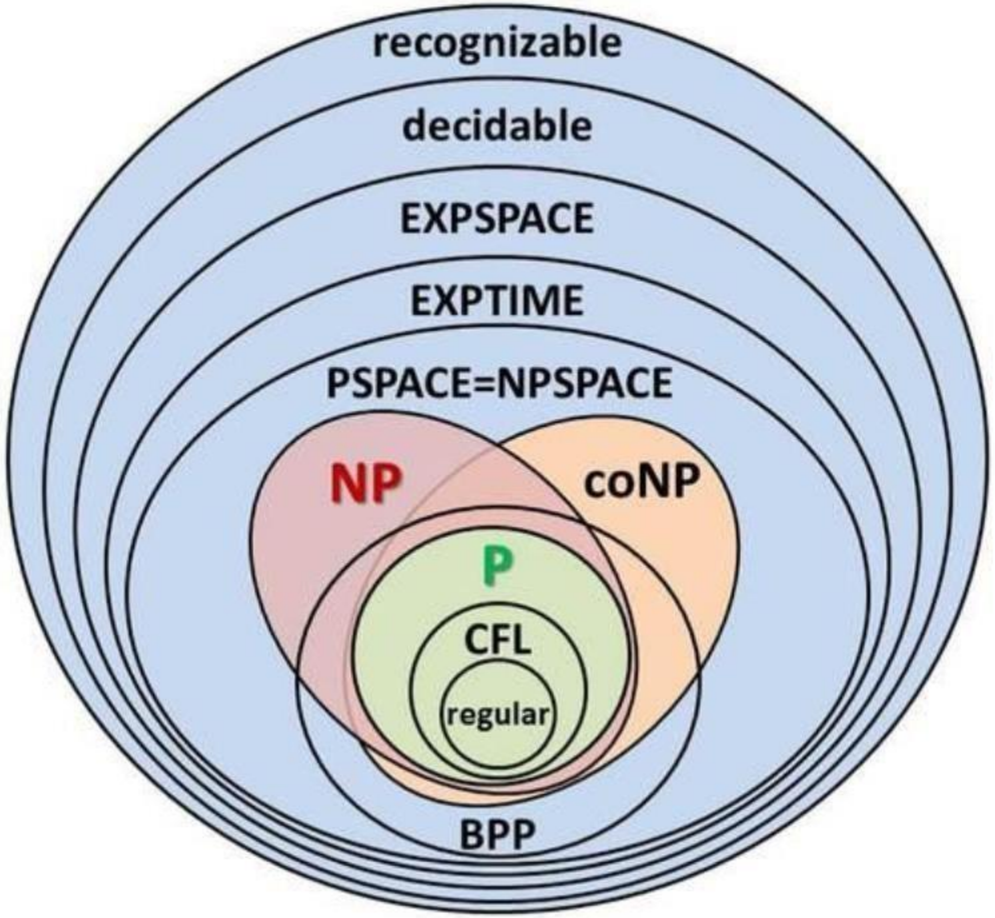
PSPACE

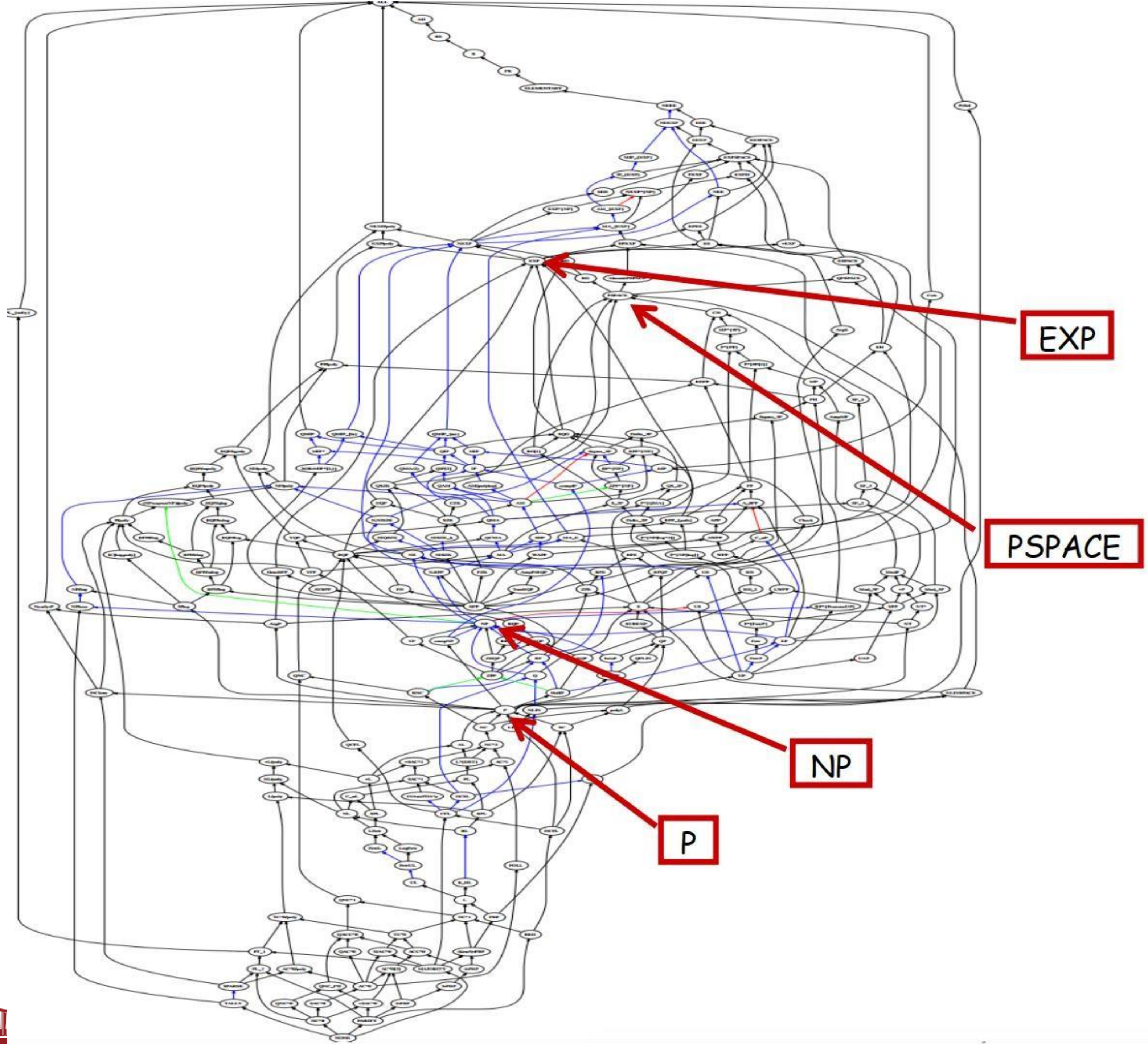
- **PSPACE.** Decision problems solvable in polynomial **space**
- **PSPACE problems**
  - QSAT
  - Planning

- **Theorem.** NP ⊆ PSPACE ⊆ EXPTIME

- **PSPACE-Complete.** Problem Y is PSPACE-complete if (i) Y is in PSPACE and (ii) for every problem X in PSPACE, $X \leq_p Y$
- **PSPACE-Complete problems**
  - QSAT
  - Competitive Facility Location

Complexity Classes


recognizable
decidable
EXPSPACE
EXPTIME
PSPACE=NPSPACE
NP
coNP
P
CFL
regular
BPP



上海科技大学
ShanghaiTech University

志成才 报国裕民